SPEECH PROCESSING

Speech signals, audio features, ASR, TTS, and evaluation

1. What is speech processing?

Speech processing is the analysis, transformation, recognition, and generation of spoken language using digital signal processing and machine learning. Audio is a signal representing changes in air pressure, captured by a microphone and converted into digital samples.

Applications include automatic speech recognition (ASR), text-to-speech (TTS), speaker recognition, voice activity detection, diarisation, speech enhancement, and spoken dialogue systems.

2. From sound to digital audio

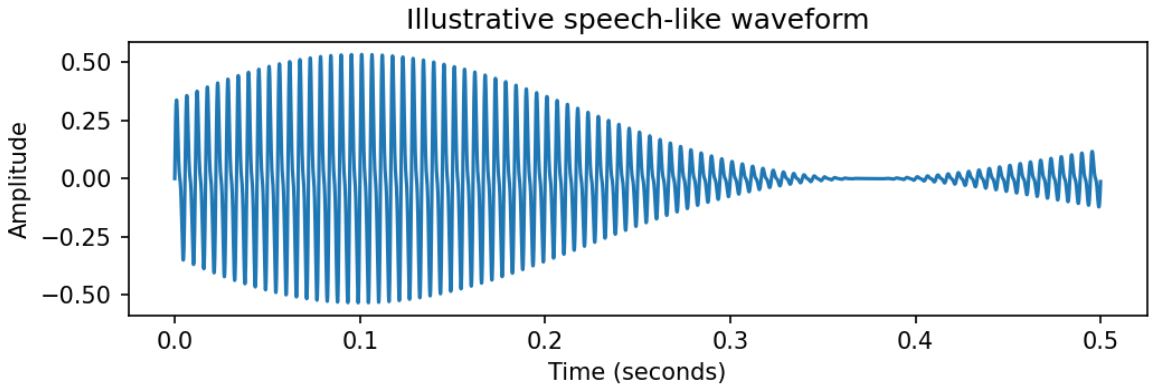
A microphone converts acoustic energy into an electrical signal. An analogue-to-digital converter samples the signal at regular intervals and quantises each sample into a numerical value. Sample rate is measured in hertz (Hz), or samples per second.

Term	Meaning	Example
Sample rate	Samples recorded per second	16,000 Hz = 16 kHz
Bit depth	Bits used to represent each sample	16-bit PCM
Channels	Separate audio streams	Mono = 1; stereo = 2
Amplitude	Magnitude of a sample	Often normalised
Duration	Length of recording	10 seconds

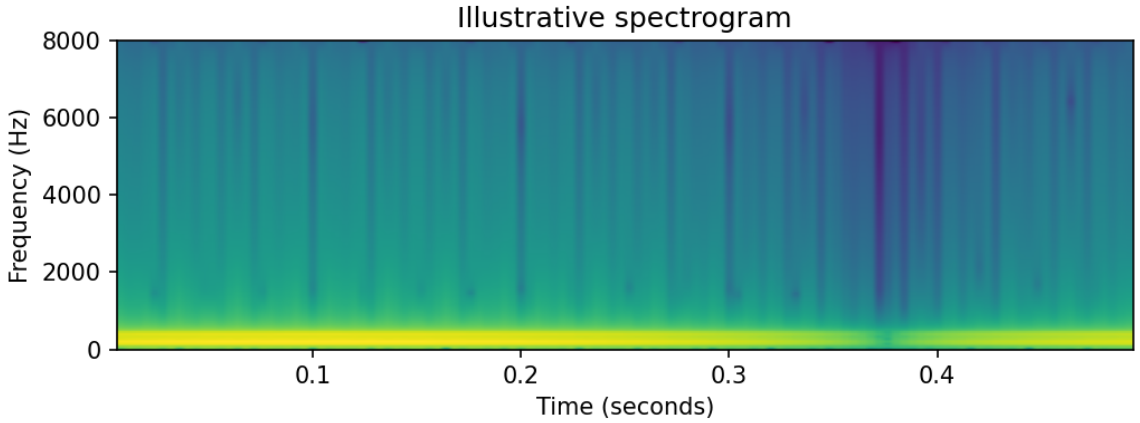
For uncompressed PCM audio, approximate bitrate is sample rate × bit depth × channel count. For example, 16 kHz, 16-bit, mono audio is about 256 kilobits per second, before file-container overhead.

3. Waveforms and spectrograms

A waveform plots amplitude over time. It can reveal pauses, clipping, and broad loudness patterns. A spectrogram represents how signal energy is distributed across frequencies over time, commonly using the Short-Time Fourier Transform (STFT).



Synthetic illustrative signal, not a human voice recording.



Illustrative spectrogram of the synthetic signal above.

4. Framing and windowing

Speech changes over time. For analysis, it is divided into short overlapping frames, often around 20–30 milliseconds, with a smaller hop between frames. A window such as the Hamming window reduces edge effects before spectral analysis.

5. Common speech features

Feature	Description	Typical use
Energy / RMS	Signal strength over a frame	Voice activity and loudness
Zero-crossing rate	Frequency of sign changes	Simple voicing/noise cues
Spectrogram	Time-frequency energy	Visual analysis and modelling
Mel spectrogram	Energy on a perceptual Mel scale	Speech and audio neural networks
MFCC	Compact cepstral features from Mel energies	Traditional speech and speaker tasks
Pitch / F0	Estimated fundamental frequency	Prosody and speaker analysis

6. Automatic Speech Recognition (ASR)

ASR converts spoken audio into text. Modern systems may use an audio encoder, sequence modelling, and a decoding strategy. End-to-end models learn mappings from audio representations to text tokens.

A typical ASR workflow includes audio validation, optional resampling or noise reduction, feature extraction or audio encoding, model inference, decoding, and transcript evaluation.

7. ASR evaluation

Word Error Rate (WER) is calculated as (substitutions + deletions + insertions) divided by the number of words in the reference transcript. Character Error Rate (CER) uses characters instead of words and can be useful when word segmentation is challenging.

Error	Definition	Example
Substitution	Reference word replaced by another	“cat” becomes “cap”
Deletion	Reference word is missing	“please call me” → “call me”
Insertion	Extra word appears in output	Output adds an unspoken “uh”

WER can exceed 100 percent when insertions are numerous. Transcript normalisation rules for casing, punctuation, numbers, and disfluencies should be consistent before scoring.

8. Text-to-Speech (TTS)

TTS converts text into generated speech. A system may normalise text, predict phonetic or acoustic representations, and generate a waveform using a vocoder or integrated neural decoder. Modern neural TTS can control speaker identity, pace, and prosody.

Component	Function
Text normalisation	Expands numbers, abbreviations, dates, and symbols into speakable forms.
Grapheme-to-phoneme	Maps written words to phonemes where pronunciation modelling is needed.
Acoustic model	Predicts acoustic features or latent representations from text.
Vocoder / decoder	Generates the audio waveform.
Prosody control	Influences pitch, rhythm, timing, and emphasis.

9. Diarisation and speaker recognition

Speaker diarisation answers “who spoke when?” by dividing a recording into speaker turns. Speaker recognition aims to identify or verify a speaker. Speaker embeddings are compact vectors representing voice characteristics and can be compared using similarity measures. Noise, channel quality, and speaking style affect both tasks.

10. Speech enhancement and VAD

Speech enhancement attempts to improve intelligibility by reducing noise, reverberation, or interference. Voice Activity Detection (VAD) identifies regions likely to contain speech. VAD errors may remove quiet speech or incorrectly retain background sounds.

11. Common tools

Tool	Typical use
FFmpeg	Audio conversion, resampling, filtering, and media inspection.
SoX	Command-line audio processing and format conversion.
librosa	Python audio analysis, feature extraction, and visualisation.
torchaudio	Audio I/O and transformations integrated with PyTorch.
Praat	Phonetic analysis, pitch, formants, and speech measurements.
Whisper	Multilingual speech recognition and transcription.

12. Audio data quality checklist

• Validate sample rate, channel count, codec, duration, and file integrity.

• Check for clipping, excessive silence, and background noise.

• Keep audio and transcript identifiers correctly aligned.

• Use consistent rules for punctuation, casing, numbers, disfluencies, and speaker labels.

• Protect personal and biometric information with appropriate access controls.

• Evaluate across accents, dialects, languages, speaking rates, and acoustic environments.

13. Key terms

ASR: Automatic Speech Recognition. **TTS:** Text-to-Speech. **STFT:** Short-Time Fourier Transform. **MFCC:** Mel-Frequency Cepstral Coefficients. **VAD:** Voice Activity Detection. **WER:** Word Error Rate. **Phoneme:** A contrastive sound unit in a language. **Prosody:** Rhythm, stress, intonation, and timing in speech.

Summary

Speech processing combines digital signal processing, linguistics, and machine learning. Understanding sampling, waveforms, spectrograms, acoustic features, ASR, TTS, diarisation, and evaluation provides a foundation for speech-enabled applications.